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WEBIGEEKS

Data Analytics With Python

Become a Job-Ready Data Analyst — From Zero to Industry-Ready in 6–7 Months

Placement Assistance

100% Job Support

Course Duration

6–7 Months

Curriculum

10+ Modules

Expert Trainers

Industry Professionals

Real-World Projects

Online & Offline Batches

Certificate on Completion

Enrol Now

Call Us

Your Complete Data Analytics Journey

The WebiGeeks Data Analytics with Python programme is structured as a comprehensive, industry-aligned curriculum. Every module builds upon the previous, taking you from spreadsheet fundamentals to advanced visualisation and NoSQL databases — everything a modern data analyst needs to be job-ready.



Excel & Data Fundamentals

Spreadsheet mastery, formulas, PivotTables, and AI-powered workflows



SQL & Relational Databases

Queries, joins, window functions, CTEs, and advanced SQL techniques



Python, Pandas & EDA

Core programming, data handling, cleaning, and exploratory analysis



Power BI, Tableau & MongoDB

BI tools, interactive dashboards, storytelling, and NoSQL databases

Excel: Interface, Data & Core Formulas

Getting Started with Excel

Navigate the Excel interface confidently — understand workbooks, worksheets, cells, rows, columns, and named ranges. Master cell referencing (relative, absolute, mixed) as the foundation for all formula work.

Essential Formula Categories

- **Mathematical:** SUM, SUMIF, SUMPRODUCT, ROUND, MOD
- **Statistical:** AVERAGE, COUNT, COUNTA, MAX, MIN, MEDIAN
- **Text Functions:** LEFT, RIGHT, MID, LEN, TRIM, CONCATENATE, TEXT
- **Logical:** IF, AND, OR, NOT, IFS, SWITCH



Excel: Advanced Lookups, Arrays & Data Quality

Lookup & Reference

VLOOKUP, HLOOKUP for simple lookups. INDEX + MATCH for dynamic, flexible references. Understand when to use each approach and their limitations in real datasets.

Array Formulas

Harness the power of array formulas for multi-condition calculations. Learn CSE arrays and the modern dynamic array functions (FILTER, SORT, UNIQUE, XLOOKUP).

Data Cleaning

Handle missing data gracefully, remove duplicates, correct errors, and use Text-to-Columns for splitting delimited data. Apply conditional formatting to flag anomalies instantly.

Data Formatting

Apply number formats, date formats, and custom formats. Use named ranges and data validation to create cleaner, error-resistant spreadsheets for professional use.

Excel: Charts, PivotTables & AI Productivity

Charts & Data Presentation

Create compelling charts — bar, column, line, pie, scatter, and combo charts. Apply best practices for data storytelling: correct chart types, clean formatting, and audience-appropriate labelling.

PivotTables & PivotCharts

Build dynamic PivotTables to summarise large datasets instantly. Create PivotCharts for interactive reporting. Customise layouts, groupings, calculated fields, and slicers for executive dashboards.

Productivity & AI Integration

- Master 50+ keyboard shortcuts to work at speed
- Use Flash Fill and AutoFill intelligently
- Leverage Excel's built-in AI: Ideas, Copilot features
- Automate repetitive tasks with Macros (introduction)
- Use Power Query for automated data import and transformation



SQL: Foundations & Database Concepts

Structured Query Language (SQL) is the universal language of data. From retrieving records to transforming complex datasets, SQL is an indispensable skill for every data analyst. This module begins with relational database theory and progresses to production-level query writing.

SQL Overview

What SQL is, its history, and its applications in analytics, reporting, and data engineering pipelines.

Relational Databases

Tables, primary keys, foreign keys, normalisation (1NF–3NF), and entity-relationship diagrams.

DDL Commands

CREATE TABLE, ALTER TABLE, DROP TABLE. Data types (INT, VARCHAR, DATE, FLOAT) and constraints (NOT NULL, UNIQUE, DEFAULT, CHECK).



SQL: Querying, Filtering & Aggregation

Core DML — SELECT Statements

Master the SELECT statement in all its forms. Filter rows with WHERE clauses. Sort results using ORDER BY (ASC/DESC). Limit output with TOP/LIMIT. Use DISTINCT to remove duplicates from result sets.

Operators & Complex Conditions

Combine conditions with AND, OR, NOT. Use BETWEEN, IN, LIKE, IS NULL for expressive filtering. Build nested conditions for complex business logic.

Aggregation & Grouping

- SUM, AVG, COUNT, MIN, MAX for metric computation
- GROUP BY to segment data by category
- HAVING to filter aggregated results
- Combining aggregations with conditional expressions

CASE Statements

Implement conditional logic directly in SQL queries. Create derived columns, custom categorisations, and business rule transformations without leaving the database layer.

SQL: Joins, Subqueries, Window Functions & CTEs

Joins — Combining Tables

INNER JOIN returns only matching rows. LEFT/RIGHT OUTER JOIN retains unmatched rows from one side. FULL OUTER JOIN combines all rows. CROSS JOIN and SELF JOIN for advanced scenarios. Master multi-table joins for real analytical queries.

Subqueries & Nested Queries

Write subqueries in SELECT, FROM, and WHERE clauses. Use correlated subqueries for row-by-row calculations. EXISTS and NOT EXISTS for efficient existence checks.

Window Functions & OVER Clause

ROW_NUMBER, RANK, DENSE_RANK for ranking. LAG, LEAD for time-series comparisons. Running totals and moving averages with SUM OVER and AVG OVER. PARTITION BY for segmented calculations.

CTEs & Recursive Queries

Common Table Expressions (WITH clause) for readable, modular queries. Multiple CTEs in a single query. Recursive CTEs for hierarchical data — org charts, bill of materials, tree structures.

Python for Data Analytics: Core Programming

Why Python for Analytics?

Python is the world's most popular language for data analysis. Its readable syntax, vast ecosystem of libraries, and community support make it the go-to tool for analysts, scientists, and engineers alike. We start with Jupyter Notebook — the industry-standard interactive environment.

Variables & Data Types

Integers, floats, strings, booleans. Type casting and type checking. String formatting with f-strings for clean output in analysis reports.

Core Data Structures

- **Lists:** Ordered, mutable sequences for collections
- **Tuples:** Immutable sequences for fixed data
- **Dictionaries:** Key-value pairs for structured records
- **Sets:** Unique element collections for deduplication

Control Flow & Functions

if/elif/else conditionals. for and while loops. List comprehensions for concise transformations. User-defined functions, scope, and return values. map(), filter(), and lambda expressions for functional-style data processing.

Pandas: Data Handling, Cleaning & Preprocessing



Reading & Writing Data

Import CSV, Excel, JSON, and SQL data into DataFrames. Export cleaned data back to multiple formats for downstream consumption.



Core DataFrame Operations

Select columns and rows. Filter with boolean masks. Sort, rank, and reshape. Apply functions across rows and columns using `.apply()` and `.transform()`.



Handling Missing Data

Detect NaN values with `.isnull()`. Fill with mean, median, or forward fill. Drop rows/columns strategically. Understand the impact of imputation choices on analysis quality.



Cleaning & Outlier Treatment

Remove duplicates with `.drop_duplicates()`. Detect and handle outliers using IQR and z-score methods. Standardise column names and fix inconsistent categorical values.



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Python Data Wrangling

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9	15	18	14	18	18	16	15	21	
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Exploratory Data Analysis & Matplotlib

Exploratory Data Analysis (EDA) is the process of visually and statistically examining datasets to uncover patterns, detect anomalies, and form hypotheses before modelling. It is one of the most critical and creative skills in a data analyst's toolkit.

Descriptive Statistics & Summarisation

Use `.describe()` for five-number summaries. Compute mean, median, mode, variance, standard deviation, skewness, and kurtosis. Understand data distribution shapes and their analytical implications.

Grouping & Aggregation

GroupBy operations for segmented analysis. Pivot tables in Pandas. SQL-like operations using `merge`, `join`, and `reshape` functions (`melt`, `pivot`, `stack`, `unstack`).

Matplotlib Visualisations

- **Line Charts:** Trends over time
- **Scatter Plots:** Correlation and distribution
- **Histograms:** Frequency distributions
- **Bar Charts:** Categorical comparisons
- **Box Plots:** Outlier detection visually

Visualisation Styling

Apply themes, colour palettes, axis labels, titles, legends, and annotations. Export publication-quality figures for reports and presentations.

NumPy, Seaborn & Statistical Analysis

Feature1	0.12	-0.12 0.47	-0.47	0.2	-0.32 -0.35
Feature2			-0.28	0.3	-0.35
Feature3	0.18	-0.12 -0.83	-0.12 0.43	0.3	-0.83 -0.33
Feature2	0.12	-0.32 -0.47	-0.12 0.47	-0.10 0.87	-0.3 -0.045
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Feature3	0.17	0.13 -0.28	-0.35	-0.12 -0.25	-0.85 0.35
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NumPy — Numerical Computing

Create and manipulate n-dimensional arrays and matrices. Vectorised operations for lightning-fast computation. Broadcasting, slicing, reshaping, and linear algebra essentials used in data pipelines and ML preprocessing.

Seaborn — Advanced Visualisation

Build statistically rich charts: heatmaps for correlation matrices, pair plots for multivariate relationships, violin plots, swarm plots, and regression plots. Apply built-in themes for publication-ready aesthetics with minimal code.

Statistical Analysis with SciPy

Fundamentals of probability distributions and inferential statistics. Hypothesis testing workflow: null hypothesis, p-values, significance levels. Implement t-tests, chi-square tests, ANOVA, and correlation tests using SciPy to validate business insights with statistical rigour.

Real-World Projects & Portfolio Building

Theory becomes expertise through practice. The WebiGeeks curriculum culminates in hands-on projects that mirror actual industry scenarios – the same kinds of assignments that hiring managers look for in a strong portfolio.

1

End-to-End Data Analysis Project

Collect, clean, analyse, and visualise a real-world dataset. Deliver a complete analytical report with findings and recommendations – from raw CSV to executive-ready insights.

2

Business Case Studies

Work through industry case studies from retail, finance, healthcare, and e-commerce. Apply the full analytics stack – SQL queries, Python analysis, and BI dashboards – to solve realistic business problems.

3

Portfolio Development

Publish projects to GitHub. Build a professional analyst portfolio with documented notebooks, dashboards, and write-ups that demonstrate competency to recruiters and hiring panels.

Power BI: From Data to Decisions

Microsoft Power BI is the world's leading business intelligence platform. This module takes you from interface basics to enterprise-grade dashboards — a skill set in high demand across every major industry.



Connect & Import

Connect to Excel, SQL, Web, and APIs. Power Query Editor for transformation.



Model & Relate

Build star schemas. Define relationships, cardinality, and cross-filter directions.



Calculate with DAX

Calculated columns, measures, CALCULATE, FILTER, time intelligence functions.



Visualise & Share

Build interactive dashboards. Publish to Power BI Service for organisational sharing.

Power BI: Advanced Features & Storytelling

Interactivity & Navigation

Drill-through pages for granular exploration. Drill-down on hierarchies. Cross-filtering and cross-highlighting between visuals. Bookmarks for guided data stories and navigation panels. Buttons and page-level actions for app-like experiences.

Real-Time & Forecasting

Connect streaming datasets for live dashboards. Use built-in forecasting on line charts. Apply clustering and anomaly detection using Power BI's built-in AI visuals.

Advanced Visualisations

- Custom visuals from the AppSource marketplace
- Decomposition Tree for root cause analysis
- Key Influencers visual for AI-driven insights
- Smart Narrative for auto-generated text summaries

Data Storytelling

Structure dashboards to guide decision-makers through a narrative. Apply layout best practices: consistent colours, clear hierarchies, and meaningful KPI cards. Design reports that answer the right business questions at a glance.

Tableau: Interactive Data Storytelling

Introduction & Interface

Tableau Desktop overview, the Tableau workspace, shelves, and cards. Understand the fundamental difference between dimensions and measures, discrete and continuous fields.

Data Connections

Connect to Excel, CSV, SQL databases, and cloud sources. Live vs. extract connections. Joins and unions within Tableau's data source layer. Data blending across multiple sources.

Dashboard Creation & Interactivity

Build multi-sheet dashboards with layout containers. Add actions — filter actions, highlight actions, URL actions. Create parameters and calculated fields for user-driven exploration.

Filters, Parameters & Data Storytelling

Context filters, dimension filters, measure filters. LOD (Level of Detail) expressions for complex aggregations. Tableau Story Points for presenting a guided analytical narrative to stakeholders.

MongoDB: NoSQL for Modern Analytics

Not all data lives in tables. MongoDB is the world's most popular NoSQL database, storing data as flexible JSON-like documents. Understanding NoSQL is increasingly essential as organisations work with semi-structured, high-volume, and rapidly changing datasets.

NoSQL Fundamentals

Why NoSQL exists alongside relational databases. Document model vs. tabular model. When to choose MongoDB over SQL — high velocity, variable schema, and unstructured data use cases.

Collections & Documents

MongoDB databases contain collections (like tables) of documents (like JSON records). Schema flexibility means each document can have different fields — ideal for evolving data models.

CRUD Operations

insertOne, insertMany, find, findOne with query operators (\$eq, \$gt, \$in). updateOne, updateMany with \$set, \$push. deleteOne, deleteMany. Aggregation pipelines for analytics.

Real-World Analytics Use

Connect MongoDB to Python using PyMongo. Run aggregation pipelines for analytical queries. Export data to Pandas DataFrames for further analysis and visualisation.

Career Opportunities After This Programme

The WebiGeeks Data Analytics with Python curriculum is designed around what employers actually hire for. Graduates are equipped to pursue multiple high-growth roles in the data space from day one.



Data Analyst

Collect, clean, and analyse structured data. Build dashboards and reports using Power BI or Tableau. Translate numbers into business insights that drive decisions across operations, sales, and marketing.



Business Analyst

Bridge the gap between data and business strategy. Use analytical skills to evaluate processes, identify opportunities, and present findings to senior stakeholders through compelling data-driven narratives.



Junior Data Scientist

Leverage the Python and statistical foundations from this programme to step into entry-level data science roles. Work on predictive models, feature engineering, and automated analytics pipelines.

Why Choose WebiGeeks?

WebiGeeks is not just a training institute — it is a career launchpad. Every aspect of the programme has been designed with one goal: making you genuinely job-ready in the shortest possible time.



Industry-Aligned Curriculum

Built in consultation with working data professionals. Every topic, tool, and project reflects what employers actually expect from analysts joining their teams today.



Live Projects & Case Studies

Learn by doing. Work on real datasets and business scenarios throughout the programme. Graduate with a portfolio that speaks louder than any certificate.



Placement Support

Dedicated placement assistance including resume building, mock interviews, LinkedIn profile optimisation, and direct connections to our industry hiring network.



Expert Trainers

Learn from practitioners with years of hands-on industry experience. Our trainers bring real-world context to every concept, not just textbook theory.

ENROL NOW

Start Your Data Analytics Journey Today

📌 Limited Seats Available — Enrol Now to Secure Your Spot

Phone

+91 8766367815

Call or WhatsApp us for admissions, queries, and batch schedules.

Email

webigeeksofficial@gmail.com

Send us your details and we'll get back to you within 24 hours.

Location

[Find Us on Google Maps](#)

Visit our centre for a free demo class. Both Online & Offline batches available.

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